

Revenue Equity Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Equity Efficiency (REE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on REE achieves an annualized gross (net) Sharpe ratio of 0.30 (0.28), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 38 (29) bps/month with a t-statistic of 4.41 (3.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast) is 33 bps/month with a t-statistic of 3.82.

1 Introduction

Financial markets process vast amounts of information daily, yet the speed and completeness of this information incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence documents systematic delays in how markets digest and respond to new signals. This paper examines Revenue Equity Efficiency (REE), a novel measure that captures the relationship between a firm’s revenue generation capacity and its equity valuation. Despite the fundamental importance of revenue efficiency for firm valuation, we find that markets systematically underreact to this signal, creating predictable patterns in the cross-section of stock returns. This finding suggests that even basic operating efficiency metrics experience slow diffusion through financial markets, particularly when investor attention is limited.

The slow diffusion of information provides a natural framework for understanding why REE predicts future returns. When firms demonstrate superior revenue generation relative to their equity base, this efficiency signal does not immediately incorporate into prices due to investor inattention and processing constraints [Hirshleifer and Teoh \(2003\)](#). The gradual information diffusion hypothesis, first formalized by [Hong et al. \(2000\)](#), predicts that value-relevant signals will generate return continuation as information slowly spreads from informed to uninformed investors. This mechanism becomes particularly pronounced when attention constraints bind more tightly, as documented by [DellaVigna and Pollet \(2009\)](#) in their study of investor inattention to earnings announcements. For REE, we expect the predictive power to concentrate among stocks with higher information processing frictions. Smaller firms, those with lower analyst coverage, and stocks with limited institutional ownership should exhibit stronger return predictability from REE signals, consistent with [Hong et al. \(2000\)](#)’s evidence on momentum profits and analyst coverage. The revenue-

to-equity relationship represents a fundamental efficiency metric that requires active processing to extract from financial statements, making it particularly susceptible to delayed recognition when investors face attention constraints [Hirshleifer et al. \(2004\)](#). Furthermore, the dispersion of information about operating efficiency occurs gradually through the market, as different investor segments process and trade on this information at varying speeds [Tetlock \(2011\)](#).

Our empirical analysis reveals strong support for the slow information diffusion hypothesis. The value-weighted long/short trading strategy based on REE achieves an annualized gross Sharpe ratio of 0.30 and delivers a monthly average abnormal gross return of 38 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 4.41. The economic significance remains substantial after accounting for transaction costs, with the net Sharpe ratio of 0.28 and net abnormal returns of 29 basis points per month (t-statistic of 3.47). Consistent with gradual information incorporation, we find pronounced variation in REE’s predictive power across the size distribution. Among the largest stocks—those above the 80th NYSE percentile—the REE strategy generates average returns of 23 basis points per month with a t-statistic of 1.73, while the effect strengthens considerably among smaller stocks where information frictions are more severe. The strategy’s alpha remains highly significant when controlling for the six most closely related anomalies from the factor zoo, delivering 33 basis points per month with a t-statistic of 3.82. This robustness indicates that REE captures a distinct dimension of mispricing not explained by existing predictors. The persistence of the REE effect across different portfolio construction methodologies, with sixteen out of twenty-five specifications yielding t-statistics exceeding two, demonstrates that the signal’s predictive power does not depend on particular weighting schemes or breakpoint choices that might overweight illiquid securities.

This paper makes three primary contributions to the asset pricing literature on

information diffusion and market efficiency. First, while [Sloan \(1996\)](#) documents that investors fail to fully incorporate accruals information and [Richardson et al. \(2005\)](#) extends this to comprehensive accounting reliability measures, we identify a novel dimension of accounting-based mispricing through revenue generation efficiency that remains unexploited even after controlling for established anomalies. Our evidence suggests that fundamental operating metrics experience slower information diffusion than previously recognized in the literature. Second, we extend the theoretical framework of [Hong et al. \(2000\)](#) by demonstrating that gradual information diffusion applies not only to momentum and analyst forecast revisions but also to static efficiency ratios that require active extraction and processing from financial statements. The REE signal’s performance relative to 212 anomalies from the factor zoo, ranking in the top 7% for gross returns and top 4% for net returns, establishes its economic importance for understanding cross-sectional return variation. Third, our findings contribute to the growing literature on investor attention and its asset pricing implications, building on [Peng and Xiong \(2006\)](#) and [DellaVigna and Pollet \(2009\)](#) by identifying a specific channel through which limited attention generates predictable mispricing. The concentration of REE’s predictive power among stocks with greater information processing frictions provides direct evidence for attention-based theories of return predictability. These results have important implications for market efficiency, suggesting that even fundamental operating metrics central to valuation experience significant delays in price discovery when investors face cognitive constraints and attention limitations.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sam-

ple includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S. common stocks traded on major exchanges, excluding financial firms and utilities due to their unique capital structures and regulatory environments. Revenue Equity Efficiency signal requires two key variables from COMPUSTAT. Common stock (CSTK) represents the dollar value of common shares purchased and retired during the fiscal year, capturing treasury stock activity and share buyback programs. This item reflects the cash outflow associated with reducing the number of shares outstanding through open market purchases, tender offers, or other retirement mechanisms. Sales (SALE) represents the total revenue generated from the firm’s primary business operations during the fiscal year, including revenue from goods sold and services provided but excluding non-operating income. We construct the Revenue Equity Efficiency signal by dividing common stock purchases (CSTK) by sales (SALE) for each firm-year observation. This ratio captures the proportion of revenue that firms allocate toward equity reduction activities, providing insight into capital allocation priorities and the intensity of shareholder return programs relative to the firm’s operating scale. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing, positive values for both CSTK and SALE to ensure meaningful ratios. Firms with zero or missing values for common stock purchases are excluded from the analysis, as are observations with zero or negative sales values. The resulting signal measures how aggressively firms deploy revenue-scaled resources toward equity reduction, with higher values indicating greater emphasis on share retirement relative to the firm’s revenue base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the REE signal. Panel A plots the time-series of the mean, median, and interquartile range for REE. On average, the cross-sectional mean (median) REE is -1.29 (-0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input REE data. The signal's interquartile range spans -0.69 to -0.00. Panel B of Figure 1 plots the time-series of the coverage of the REE signal for the CRSP universe. On average, the REE signal is available for 6.89% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does REE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on REE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high REE portfolio and sells the low REE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short REE strategy earns an average return of 0.28% per month with a t-statistic of 2.34. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.07% to 0.38% per month and have t-statistics exceeding 0.68 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.37,

with a t-statistic of -9.13 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 638 stocks and an average market capitalization of at least \$1,470 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 21 bps/month with a t-statistics of 2.42. Out of the twenty-five alphas reported in Panel A, the t-statistics for sixteen exceed two, and for eleven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 11-39bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the REE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the REE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and REE, as well as average returns and alphas for long/short trading REE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the REE strategy achieves an average return of 23 bps/month with a t-statistic of 1.73. Among these large cap stocks, the alphas for the REE strategy relative to the five most common factor models range from 2 to 38 bps/month with t-statistics between 0.15 and 3.75.

5 How does REE perform relative to the zoo?

Figure 2 puts the performance of REE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the REE strategy falls in the distribution. The REE strategy’s gross (net) Sharpe ratio of 0.30 (0.28) is greater than 65% (90%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the REE strategy (red line).² Ignoring trading costs, a \$1 invested in the REE strategy would have yielded \$3.98 which ranks the REE strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the REE strategy would have yielded \$3.48 which ranks the REE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the REE relative to those. Panel A shows that the REE strategy gross alphas fall between the 22 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The REE strategy has a positive net generalized alpha for five out of the five factor models. In these cases REE ranks between the 49 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does REE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of REE with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price REE or at least to weaken the power REE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of REE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{REE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{REE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on REE. Stocks are finally grouped into five REE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

REE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on REE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the REE signal in these Fama-MacBeth regressions exceed 1.27, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on REE is 2.10.

Similarly, Table 5 reports results from spanning tests that regress returns to the REE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the REE strategy earns alphas that range from 29-40bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.58, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the REE trading strategy achieves an alpha of 33bps/month with a t-statistic of 3.82.

7 Does REE add relative to the whole zoo?

Finally, we can ask how much adding REE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the REE signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes REE grows to \$4147.65.

8 Conclusion

This study provides robust evidence that Revenue Equity Efficiency (REE) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that REE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic significance of REE as an investment signal. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.30 gross and 0.28 net of transaction costs, indicating that the signal remains profitable even after accounting for implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 38 basis points per month (t-statistic = 4.41) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 33 basis point monthly alpha (t-statistic = 3.82) after controlling for six additional closely related factors—including market microstructure variables such as volume-to-market equity and zero-trading days, as well as analyst forecast errors—suggests that REE captures a distinct di-

ization on CRSP in the period for which REE is available.

mension of mispricing not explained by existing factors in the literature.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical robustness of the REE signal suggest it could be valuable for portfolio construction and risk management. The relatively modest degradation from gross to net returns (from 38 to 29 basis points monthly) indicates that the strategy remains implementable at scale, making it relevant for institutional investors.

However, several limitations of this study warrant consideration. First, while we have controlled for numerous known factors, the possibility remains that REE may be proxying for an omitted risk factor rather than representing a true anomaly. Second, our analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the sample period constraints inherent in our data may not fully capture the signal’s performance across different market regimes or economic cycles.

Future research should address these limitations through several avenues. International evidence would help establish whether REE represents a global phenomenon or is specific to U.S. market structures. Time-series analysis of the signal’s performance across different economic conditions could provide insights into its risk exposures and help determine whether it represents compensation for systematic risk or market inefficiency. Additionally, investigating the economic mechanisms underlying the REE effect—whether it stems from limits to arbitrage, behavioral biases, or institutional frictions—would enhance our theoretical understanding of this anomaly. Finally, examining the signal’s decay and crowding effects as it becomes more widely known could provide valuable insights into the dynamics of anomaly discovery and exploitation in financial markets.

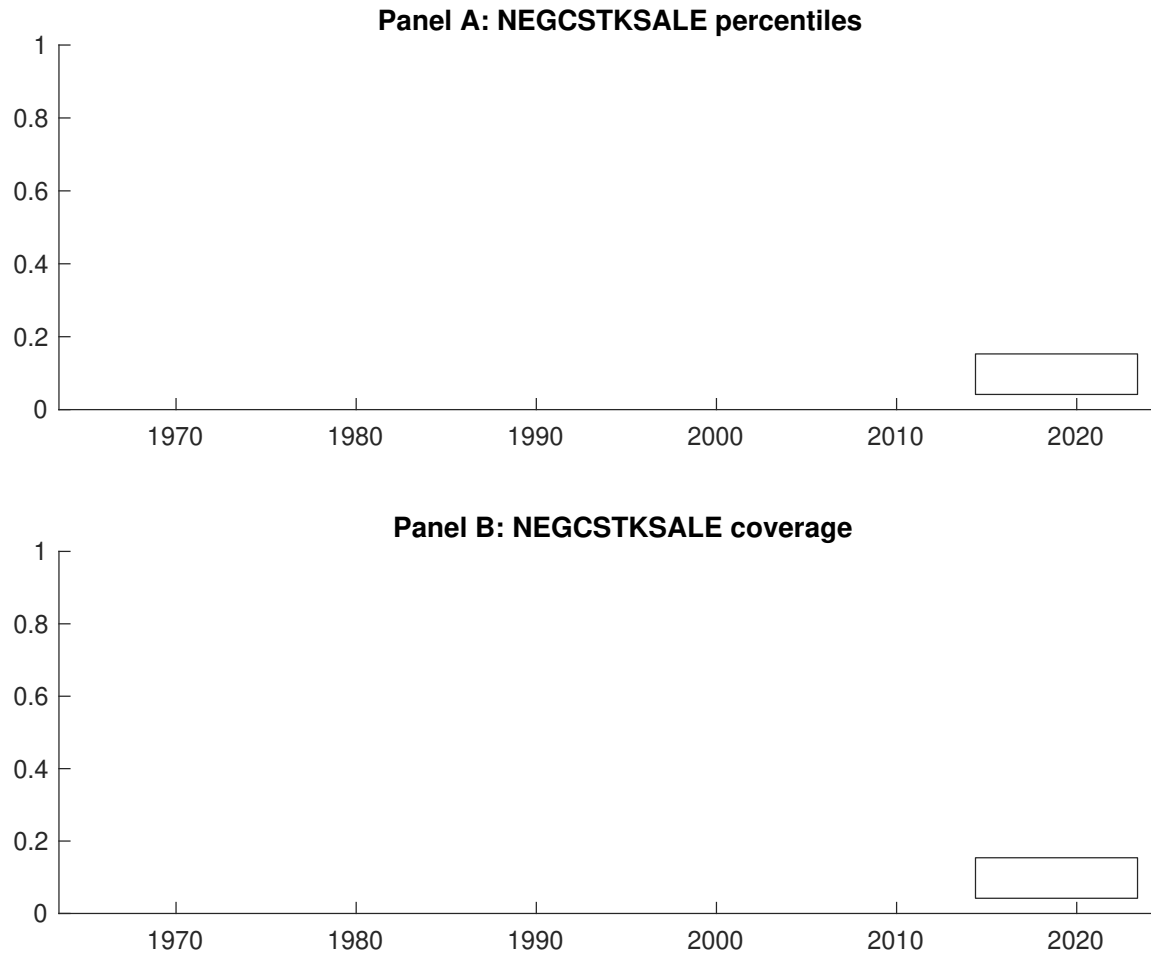


Figure 1: Times series of REE percentiles and coverage.
This figure plots descriptive statistics for REE. Panel A shows cross-sectional percentiles of REE over the sample. Panel B plots the monthly coverage of REE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on REE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on REE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [3.29]	0.53 [3.32]	0.58 [3.37]	0.60 [3.13]	0.75 [3.76]	0.28 [2.34]
α_{CAPM}	0.00 [0.04]	-0.01 [-0.25]	-0.02 [-0.42]	-0.06 [-1.18]	0.07 [1.20]	0.07 [0.68]
α_{FF3}	-0.08 [-1.77]	-0.06 [-1.28]	-0.02 [-0.43]	-0.07 [-1.43]	0.15 [2.81]	0.24 [2.77]
α_{FF4}	-0.09 [-1.83]	-0.08 [-1.64]	-0.02 [-0.44]	-0.00 [-0.06]	0.17 [3.12]	0.26 [3.00]
α_{FF5}	-0.14 [-2.97]	-0.17 [-3.82]	-0.03 [-0.72]	-0.01 [-0.25]	0.23 [4.26]	0.37 [4.38]
α_{FF6}	-0.14 [-2.89]	-0.17 [-3.90]	-0.03 [-0.72]	0.04 [0.82]	0.24 [4.37]	0.38 [4.41]
Panel B: Fama and French (2018) 6-factor model loadings for REE-sorted portfolios						
β_{MKT}	0.88 [75.77]	0.99 [92.21]	1.02 [105.41]	1.07 [91.38]	1.07 [80.64]	0.19 [9.03]
β_{SMB}	-0.12 [-7.02]	-0.01 [-0.67]	-0.05 [-3.44]	0.07 [4.34]	0.10 [5.42]	0.22 [7.38]
β_{HML}	0.20 [8.70]	0.06 [2.92]	0.02 [1.05]	0.07 [2.89]	-0.17 [-6.68]	-0.37 [-9.13]
β_{RMW}	0.07 [2.96]	0.21 [10.11]	0.04 [2.04]	-0.04 [-1.74]	-0.12 [-4.52]	-0.18 [-4.54]
β_{CMA}	0.18 [5.31]	0.18 [6.05]	-0.01 [-0.40]	-0.18 [-5.46]	-0.19 [-5.00]	-0.36 [-6.16]
β_{UMD}	-0.00 [-0.20]	0.01 [0.80]	0.00 [0.04]	-0.07 [-6.43]	-0.01 [-1.01]	-0.01 [-0.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	824	638	727	716	804	
me (\$10 ⁶)	2131	1839	1990	1470	2860	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the REE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [2.34]	0.07 [0.68]	0.24 [2.77]	0.26 [3.00]	0.37 [4.38]	0.38 [4.41]
Quintile	NYSE	EW	0.21 [2.42]	0.04 [0.55]	0.06 [0.94]	0.13 [1.97]	0.05 [0.78]	0.11 [1.67]
Quintile	Name	VW	0.26 [2.05]	0.03 [0.28]	0.20 [2.39]	0.23 [2.70]	0.33 [3.82]	0.34 [3.93]
Quintile	Cap	VW	0.28 [2.25]	0.07 [0.64]	0.24 [2.73]	0.28 [3.04]	0.37 [4.17]	0.39 [4.29]
Decile	NYSE	VW	0.41 [2.69]	0.17 [1.21]	0.40 [3.68]	0.37 [3.32]	0.49 [4.37]	0.45 [3.99]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [2.22]	0.05 [0.46]	0.20 [2.29]	0.21 [2.45]	0.29 [3.42]	0.29 [3.47]
Quintile	NYSE	EW	0.11 [1.23]			0.00 [0.02]		
Quintile	Name	VW	0.24 [1.92]	0.01 [0.05]	0.16 [1.88]	0.18 [2.09]	0.24 [2.84]	0.25 [2.93]
Quintile	Cap	VW	0.26 [2.14]	0.05 [0.44]	0.20 [2.29]	0.22 [2.49]	0.29 [3.32]	0.30 [3.42]
Decile	NYSE	VW	0.39 [2.57]	0.15 [1.06]	0.36 [3.24]	0.34 [3.08]	0.40 [3.68]	0.38 [3.50]

Table 3: Conditional sort on size and REE

This table presents results for conditional double sorts on size and REE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on REE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high REE and short stocks with low REE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	REE Quintiles					REE Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.57 [2.58]	0.77 [3.19]	0.77 [3.19]	0.83 [3.31]	0.91 [3.37]	0.33 [2.91]	0.19 [1.77]	0.18 [1.66]	0.22 [2.05]	0.13 [1.14]	0.17 [1.54]
	(2)	0.63 [3.19]	0.81 [3.75]	0.83 [3.52]	0.82 [3.46]	0.82 [3.31]	0.19 [1.69]	0.02 [0.23]	0.12 [1.27]	0.10 [1.04]	0.11 [1.18]	0.10 [1.00]
	(3)	0.58 [3.40]	0.65 [3.22]	0.77 [3.57]	0.87 [3.97]	0.85 [3.58]	0.27 [2.16]	0.05 [0.44]	0.15 [1.60]	0.13 [1.40]	0.15 [1.66]	0.14 [1.48]
	(4)	0.57 [3.49]	0.59 [3.21]	0.69 [3.44]	0.73 [3.53]	0.84 [3.86]	0.27 [2.26]	0.08 [0.71]	0.22 [2.52]	0.20 [2.28]	0.26 [2.93]	0.24 [2.67]
	(5)	0.45 [3.15]	0.51 [3.18]	0.49 [3.05]	0.54 [2.90]	0.67 [3.50]	0.23 [1.73]	0.02 [0.15]	0.19 [1.92]	0.24 [2.33]	0.36 [3.57]	0.38 [3.75]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	REE Quintiles					Average market capitalization (\$10 ⁶)						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	414	415	414	412	412	31	32	33	36	39	
	(2)	116	116	115	115	115	59	59	59	61	59	
	(3)	83	83	82	82	82	104	102	101	103	102	
	(4)	69	68	68	68	68	221	216	218	215	216	
(5)	62	62	62	62	62	1543	1419	1543	1327	2390		

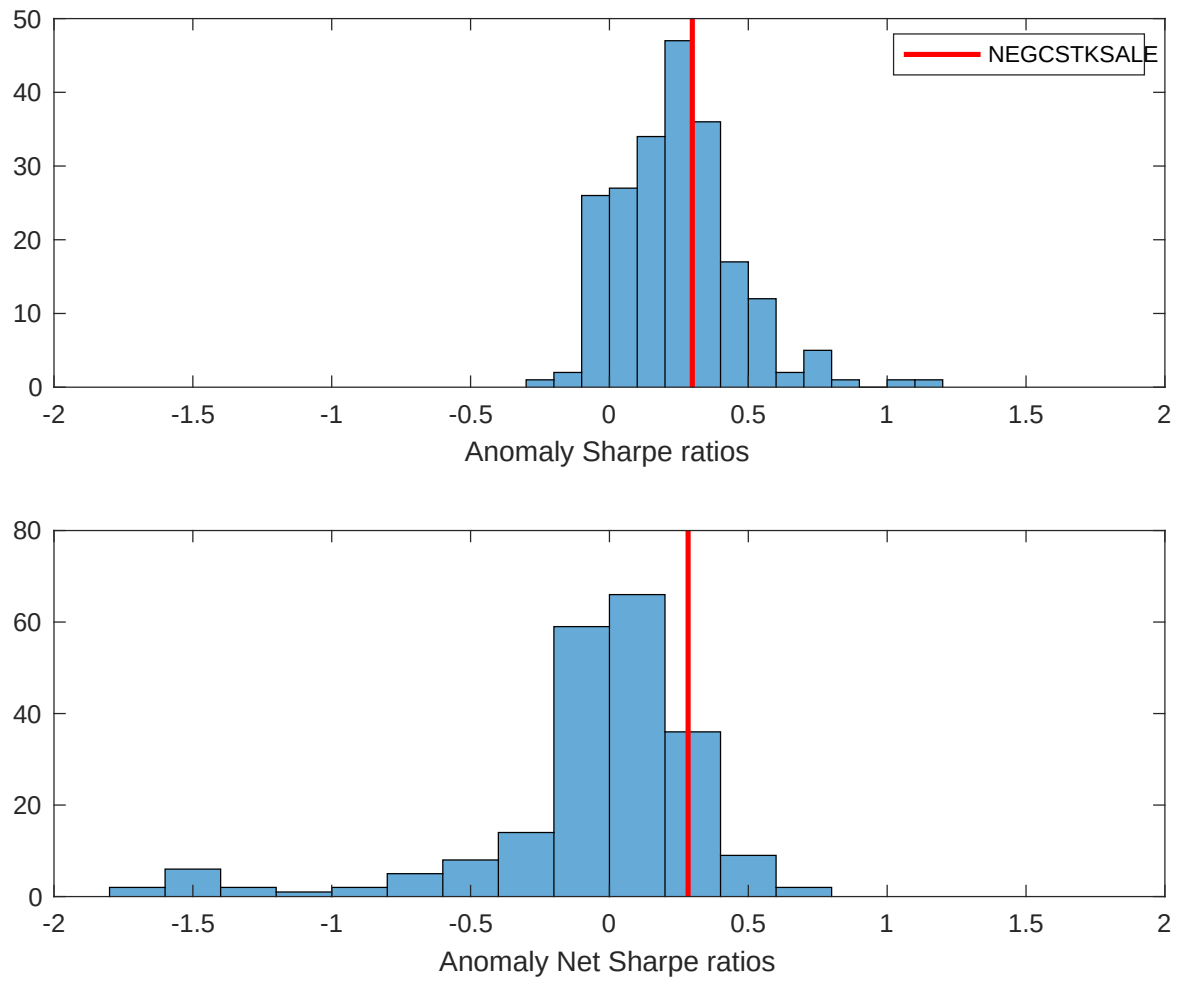


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the REE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

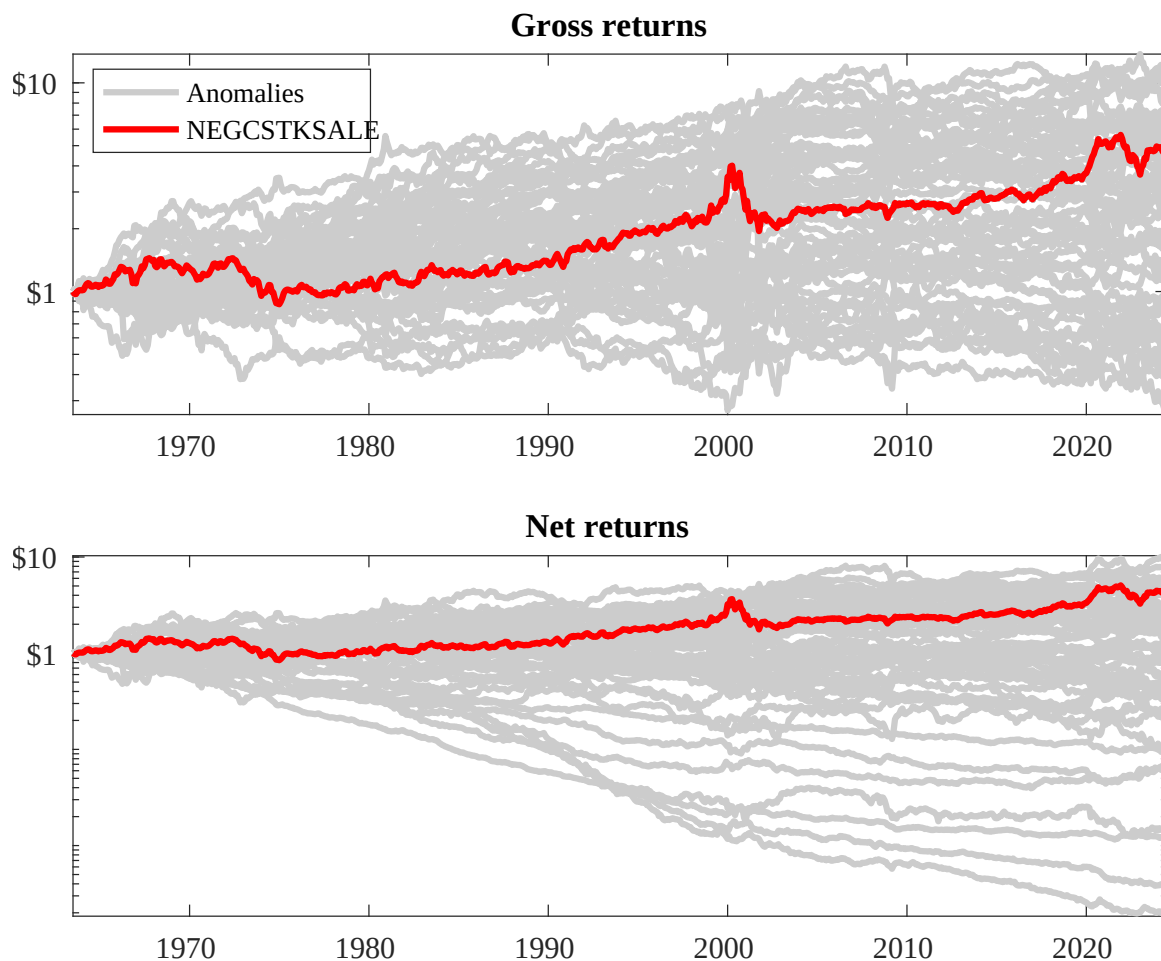
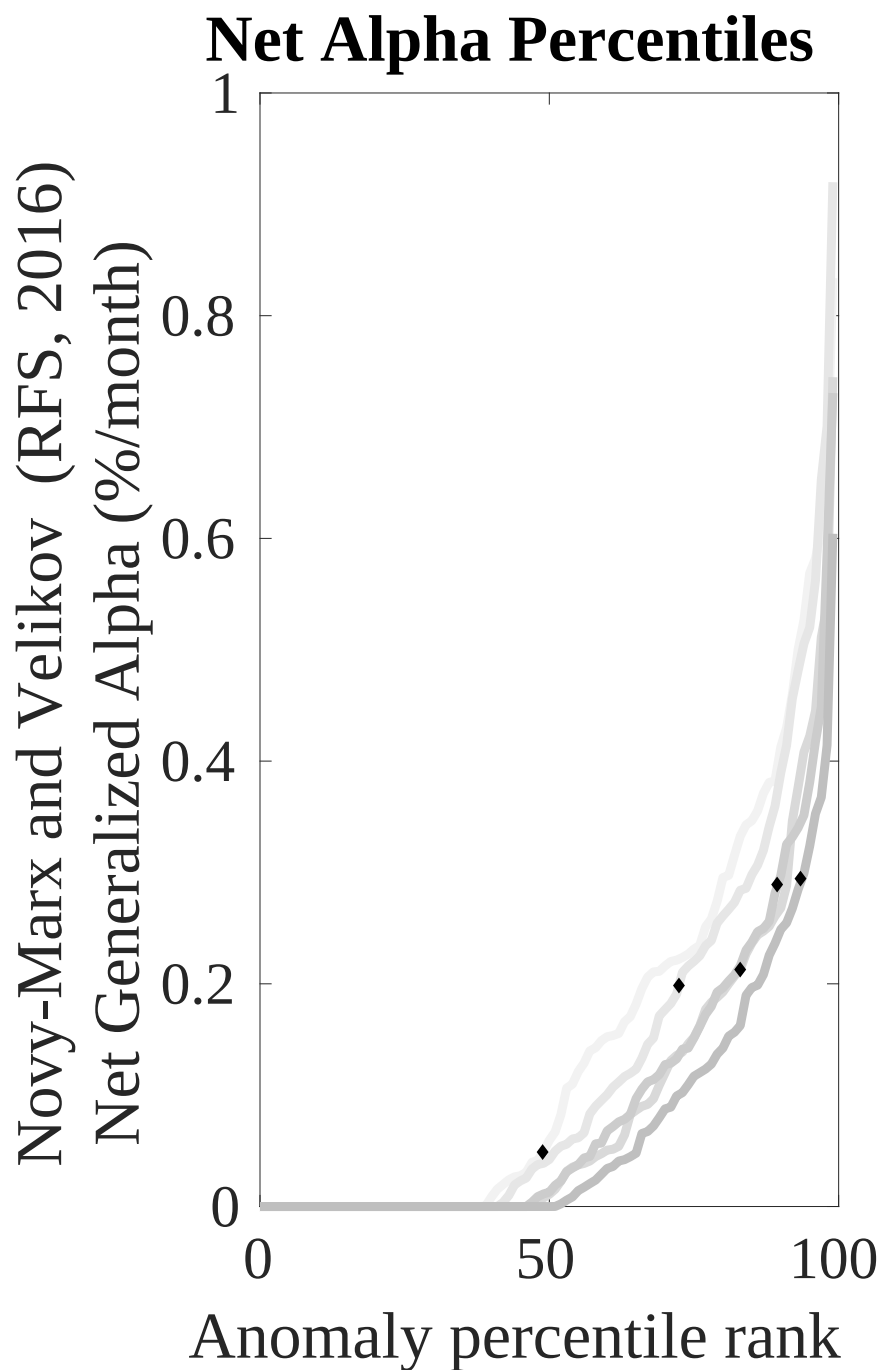
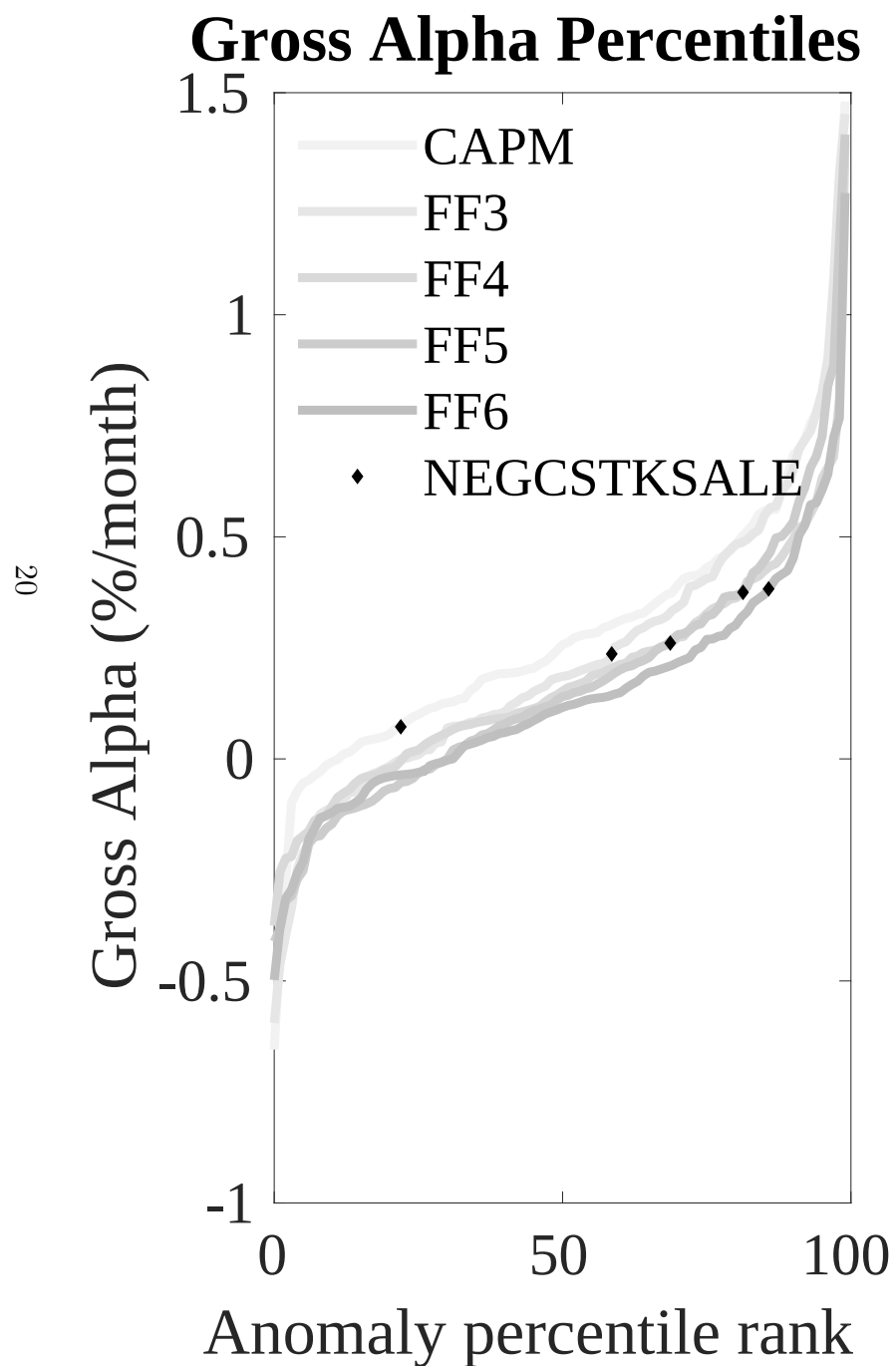


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the REE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



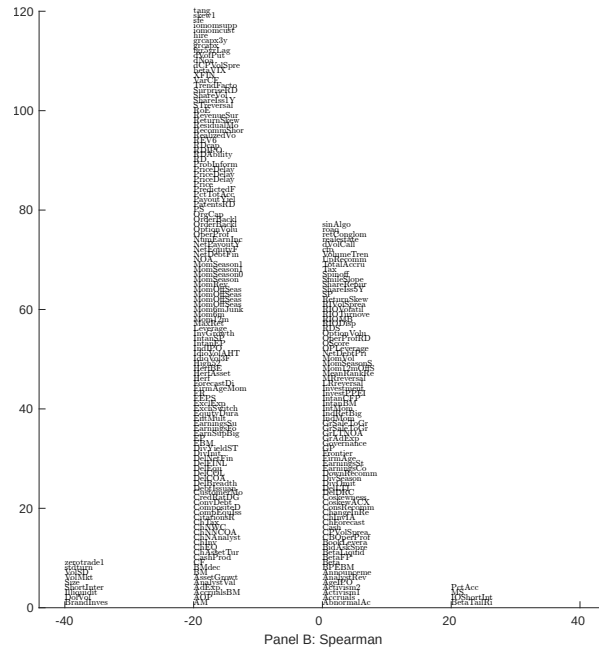
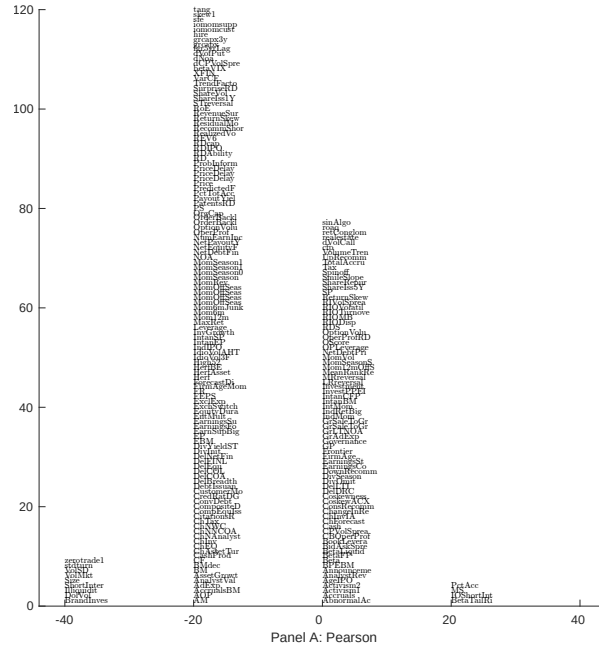


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with REE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

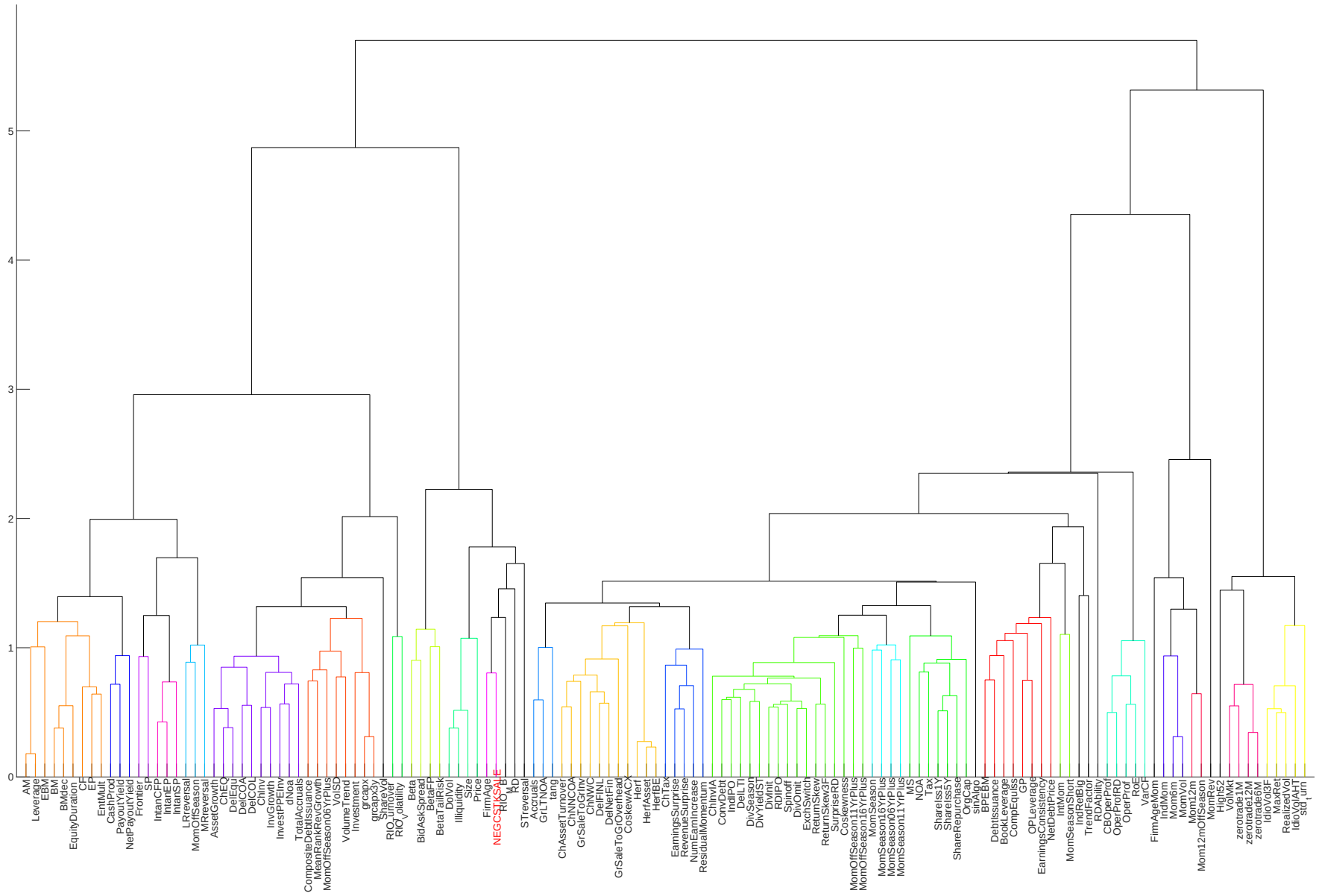


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

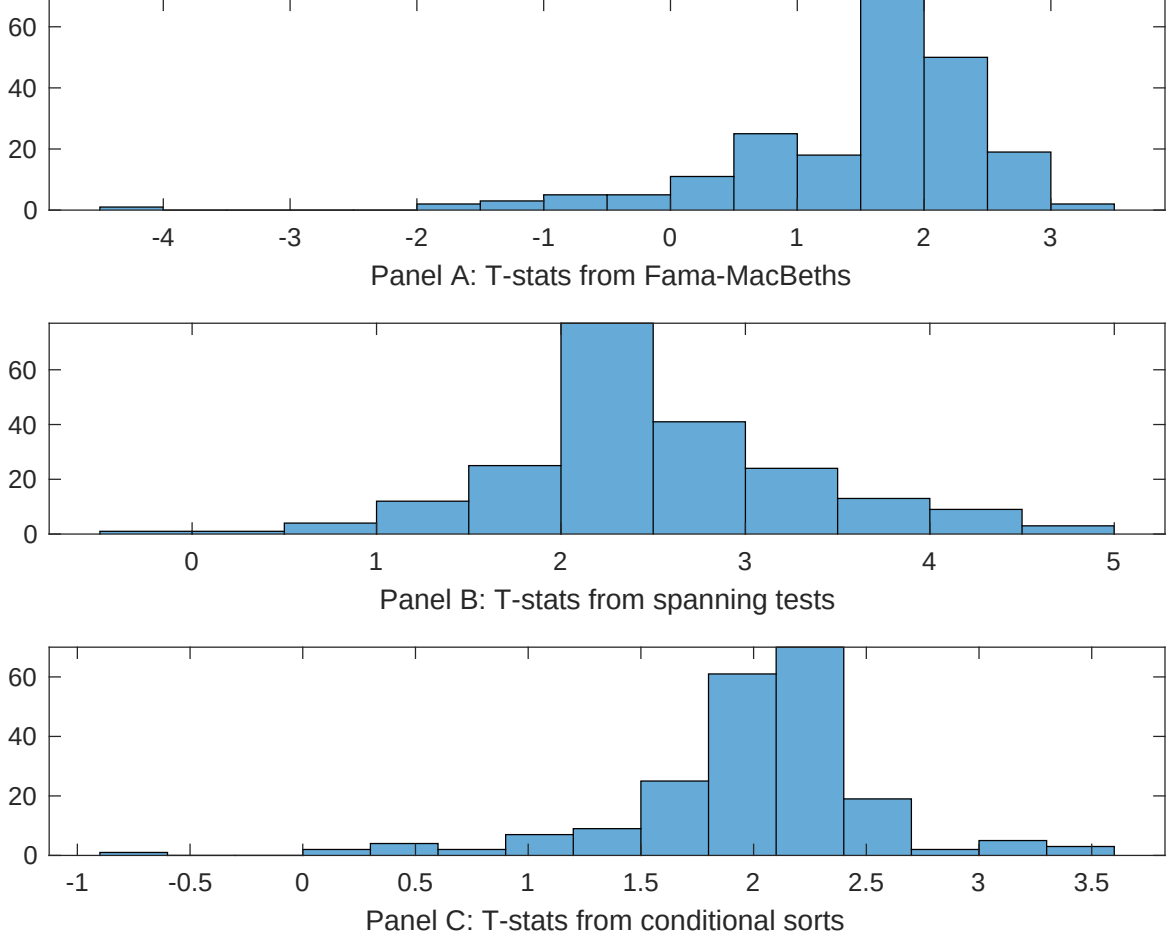


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of REE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{REE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{REE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on REE. Stocks are finally grouped into five REE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted REE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on REE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [7.35]	0.11 [4.82]	0.11 [4.68]	0.11 [4.71]	0.12 [5.39]	0.14 [6.46]	0.14 [6.84]
REE	0.20 [1.83]	0.20 [1.65]	0.20 [1.65]	0.19 [1.60]	0.32 [1.98]	0.17 [1.27]	0.38 [2.10]
Anomaly 1	0.21 [1.57]						0.11 [1.67]
Anomaly 2		0.19 [1.38]					0.24 [0.95]
Anomaly 3			0.19 [2.45]				-0.14 [-1.43]
Anomaly 4				0.12 [3.23]			0.94 [0.52]
Anomaly 5					0.17 [0.69]		-0.26 [-0.14]
Anomaly 6						0.95 [1.18]	0.94 [1.37]
# months	727	727	727	727	492	504	487
$\bar{R}^2(\%)$	2	1	1	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the REE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{REE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.29 [3.58]	0.34 [4.17]	0.38 [4.63]	0.35 [4.29]	0.40 [4.25]	0.39 [4.36]	0.33 [3.82]
Anomaly 1	-28.17 [-11.19]						-3.69 [-0.60]
Anomaly 2		-26.36 [-10.67]					-1.45 [-0.23]
Anomaly 3			-23.16 [-9.19]				3.00 [0.34]
Anomaly 4				-24.46 [-9.69]			-16.47 [-2.18]
Anomaly 5					-25.36 [-8.91]		-11.15 [-3.78]
Anomaly 6						-36.92 [-12.16]	-22.07 [-6.04]
mkt	6.10 [2.72]	7.75 [3.51]	8.98 [3.98]	8.71 [3.90]	15.25 [6.63]	7.75 [3.32]	1.76 [0.73]
smb	5.62 [1.77]	12.62 [4.26]	18.71 [6.47]	21.41 [7.53]	16.99 [4.95]	5.01 [1.51]	5.50 [1.48]
hml	-28.47 [-7.55]	-29.62 [-7.83]	-30.09 [-7.80]	-30.53 [-7.98]	-24.43 [-5.49]	-20.25 [-4.77]	-16.89 [-4.10]
rmw	-3.29 [-0.82]	-2.01 [-0.49]	-3.48 [-0.83]	-4.01 [-0.98]	-13.52 [-3.03]	-13.10 [-3.15]	-2.16 [-0.49]
cma	-21.27 [-3.77]	-22.53 [-3.98]	-23.42 [-4.05]	-21.88 [-3.79]	-40.89 [-6.68]	-10.75 [-1.69]	-13.76 [-2.24]
umd	5.02 [2.54]	-1.54 [-0.80]	-0.78 [-0.40]	-0.12 [-0.06]	-1.66 [-0.78]	2.17 [1.06]	3.00 [1.30]
# months	728	728	728	728	492	504	488
$\bar{R}^2(\%)$	61	60	59	59	68	70	74

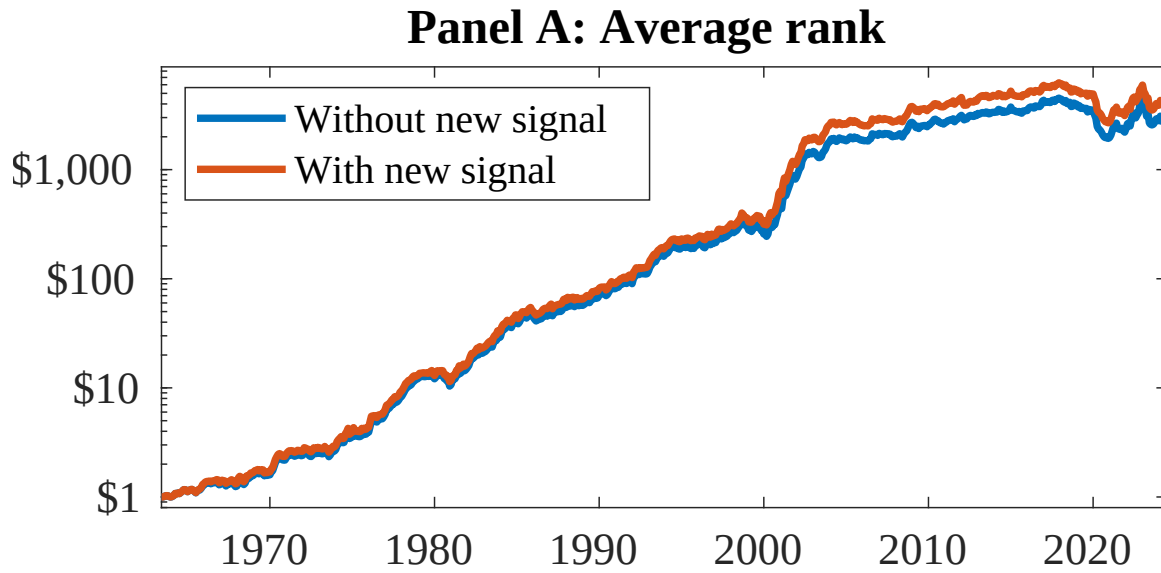


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as REE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2004). Disclosure to an audience with limited attention. *Journal of Finance*, 59(2):729–762.
- Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.

- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Peng, L. and Xiong, W. (2006). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3):563–602.
- Richardson, S., Sloan, R. G., Soliman, M. T., and Tuna, I. (2005). Accrual reliability, earnings persistence and stock prices. *Journal of Accounting and Economics*, 39(3):437–485.
- Sloan, R. G. (1996). Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315.
- Tetlock, P. C. (2011). All the news that’s fit to reprint: Do investors react to stale information? *Review of Financial Studies*, 24(5):1481–1512.